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ORIGINAL RESEARCH

T + 0 Settlement: A Game Changer for the Indian Stock Market

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Abstract

The Indian equity market is growing rapidly, supported by a diverse investor base and strong growth. The study's objective is to analyse the impact of T + 0 settlement on the cumulative abnormal returns and cumulative average abnormal returns of included stocks. The present study will also study the impact of post-T + 0 settlements on trading volumes. Using event study methodology for 2,5,10, and 20-day event windows and Dynamic Forecasting Technique, the paper analysed 25 selected stocks and observed its potential in reshaping market efficiency and liquidity. The findings indicate the semi-strong form of market efficiency reveals a mixed response across different stocks, suggesting that the impact of T + 0 settlement is not uniform. Some companies, notably **Vedanta, Indian Hotels, Tata Communications, and Hindalco** experienced significant positive abnormal returns, indicating an immediate positive reception of the shortened settlement cycle. Vedanta, in particular, showed a consistently strong positive trend across all event windows, culminating in a substantial 36% abnormal return over 20 days. However, the companies like **Coforge, Birla Soft, and MRF** experienced significant negative abnormal returns, especially in the longer event windows. This outcome disparity suggests that the benefits of T + 0 settlement may be sector-specific or dependent on individual stock characteristics. Interestingly, while individual stocks showed significant movements, the cumulative average abnormal returns (CAAR) across all sample stocks were insignificant. While the T + 0 settlement had notable effects on certain stocks, its overall market impact remains subtle at this early stage. The volume analysis provides encouraging signs for the potential of T + 0 settlement. An average increase of 3% in trading volume across the participating stocks, even with the optional nature of the current implementation, suggests the willingness of market participants to receive this new settlement cycle positively. Dynamic forecasting technique also confirms the significant differences in the patterns of the above seven stock prices during the entire 41-day event window. Therefore, these contemporary results hold major inferences for the market regulator.

Keywords T + 0 settlement cycle · stock market · Dynamic Forecasting Technique

Extended author information available on the last page of the article

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Abbreviations

FPIs Foreign Portfolio Investors
DIIs Domestic Institutional Investors
SEBI Securities and Authors 'Exchange Board of India
DLT Distributed ledger technology

1 Introduction

The Indian equity market is growing rapidly, supported by a diverse investor base and strong growth. In 2023–24, despite global uncertainties like high interest rates and inflation, optimism around economic growth, rate cuts, and declining commodity prices kept markets resilient (Dhingra et al., 2024). Foreign Portfolio Investors (FPIs) provide liquidity but remain sensitive to global shifts, while Domestic Institutional Investors, with increased retail investor participation, add stability. Retail investors, supported by digital platforms and High net worth individuals, are restructuring market dynamics. The increase in midcap and small-cap stocks, along with record retail and derivative activity, has made Securities and Exchange Board of India (SEBI) increase regulatory oversight for investor protection (Aggrawal & Pandey, 2025). To improve market efficiency, SEBI introduced key reforms, including the beta launch of T+ 0 settlement, the Investor Risk Reduction Access platform, and the Corporate Debt Market Development Fund. T+ 0 settlement, in particular, is expected to be a game-changer, enhancing liquidity, reducing risk, and boosting investor confidence. As markets hit record highs, balancing growth with risk is crucial.

Figure 1 clearly displays the stakeholders of the Indian equity market, divided into Investors, Market Intermediaries, and Market Infrastructure. Investors comprises of institutional, retail, and high-net-worth individuals. Market intermediaries comprise stock brokers, asset management companies, and custodians. Market infrastructure includes stock exchanges, clearing corporations, and depositories, all regulated by SEBI.

SEBI's introduction of the beta version of the T+ 0 settlement cycle on March 28, 2024, for 25 stocks is a global first (Jarrow & Chatterjea, 2024). The T+ 0 settlement was introduced to help control the counterparty risk and improve overall market efficiency (Colle, 2021). This study is the first to evaluate the impact of T+ 0 settlement in Indian equity markets, providing an understanding for market regulators, legislators, investors, and other market participants. The present work holds weightage due to integrating global markets and surging transaction volumes for efficient and safe settlement processes, ensuring market stability and enhanced investor confidence (Alba et al., 2002).

In 2002, the settlement cycle was reduced by SEBI from T+ 5 to T+ 3 (Kutchu, 2012). It was further reduced to T+ 2 and T+ 1 in the years 2003 and 2021, respectively. In 2023, T+ 1 was made a compulsory requirement (Gogia et al., 2023).

In the first stage, the T+ 0 settlement was applied to 25 stocks, and for the remaining stocks, the existing T+ 1 cycle was followed. This measure aims to enhance market liquidity and reduce risks with long settlement periods; T+ 0

Fig. 1 Stakeholders of the Indian Equity Markets. Source: Authors own work

settlement expands capital efficiency and reduces counterparty risk by allowing the quick transfer of securities and funds (Abel, 2022). This regulatory initiative would bring India's trading structure to global standards, increasing market depth and bringing in more investors (McLaughlin, 2023).

Until T + 1, the seller receives 80% of their cash on the transaction day, with the remaining 20% the next day. The proposed T + 0 system would allow immediate access to 100% of the settled trade value. It would be implemented in two phases: Phase 1 for transactions before 1:30 pm, settled by 4:30 pm the same day; and Phase 2 for transactions between 1:30 pm and 3:30 pm, with Phase 1 no longer in effect.

The Indian stock market structure is centralized, with government bodies, financial institutions, and the private sector depending on each other to validate transactions. Due to this reliance, settlements in client Demat accounts occur on a T + 2 basis. This T + 2 rule often leads to financial losses or missed profits for retail short-term investors or traders. Additionally, the complexity of the existing settlement process frequently leads to discrepancies in customer Demat accounts.

Around the world, many financial jurisdictions are exploring the possibility of reducing the settlement cycle for equity markets from T + 2 to T + 1. The main goals of this shift are to enhance investor protection, minimize risk, and boost efficiency. Shortening the settlement cycle also reduces the demand for funding liquidity by lowering margin requirements, potentially unlocking liquidity for less actively traded stocks. Analyzing data from the Indian markets, which transitioned securities from T + 2 to T + 1 on a monthly basis, a series of quasi-natural tests were conducted which indicated that shorter settlement cycles enhance

market liquidity, with pronounced effect on stocks with lower market capitalization. Thus, reducing settlement cycles may improve overall liquidity and extend it to fewer liquid stocks. (Bhanu et al., 2024).

In the T+ 0 system, institutional investors, particularly foreign funds, may encounter challenges. Unlike retail traders who deposit funds before trading, large funds operate differently. Immediate settlement necessitates pre-funding accounts, which expose these investors to currency risks. Brokers observe that foreign investors need to allocate funds at least a day in advance due to time zone differences involving intermediaries like custodian banks, foreign exchange banks, and brokers. When SEBI shortened the settlement cycle to T+ 1 from T+ 2 in 2023, foreign fund managers were among the most vocal critics.

Brokers serving retail clients expect increased competition, where financial management efficiency will be crucial. The shorter settlement cycle may lower their funding needs and speed up the release of funds from exchanges. However, operational challenges, particularly in trade reconciliation, are likely to be a concern during the initial phase.

This study is the first of its kind to examine the impact of T+ 0 settlement on the Indian equity market. An event study technique has been used to analyze the impact of the T+ 0 settlement in the 2, 5, 10, and 20-day event windows. This study will help to understand the market reactions and will help the changes to be incorporated in future regulatory implementation. Also, it will help to understand whether the T+ 0 settlement has helped change stock valuations or investment strategies among various stakeholders. Shortened settlement cycles can prove to be a game changer for existing market dynamics (Cassells, 2023). The study's objective is to analyse the impact of T+ 0 settlement on the cumulative abnormal returns (CAR) and cumulative average abnormal returns (CAAR) of sample stocks. The present study will also analyze the impact of post-T +0 settlements on trading volumes. This will help to highlight the reaction of investors and traders looking forward to the proposed full implementation of T+ 0 settlement cycles for portfolio creation or short-term gains. An increase in trading volume implies enhanced market liquidity and potentially lower transaction costs, whereas a fall suggests caution among market participants (Bianchi et al., 2022).

The findings of this study are likely to have efficacious consequences for various stakeholders in the financial markets. The results will also deliver pragmatic evidence to inspect the efficacy of the T+ 0 settlement cycle in fulfilling the intended objectives. This study will contribute to the literature on dynamic changes in market structural reforms with a regulatory reach. India's emergence as a pioneer in market innovation will be highlighted in this study. With the introduction of the T+ 0 settlement cycle, India will be a leader in combating the challenges of modern financial markets to set benchmarks in global financial markets. The outcome of the T+ 0 settlement may be considered as a contemporary case study for other world markets contemplating similar reforms.

Providing empirical evidence on the effects of shortened settlement cycles can lead to informed policy decisions, guide market practices, and contribute to the ongoing evolution of global financial markets towards greater efficiency and stability.

The shift to a T +0 settlement cycle pens a pioneering change in India's equity market, positioning the country as a global leader in financial market innovation. SEBI's initiative to implement same-day settlements is a first-of-its-kind development, reflecting India's proactive stance in enhancing market efficiency, reducing counterparty risks, and improving liquidity (Panda, 2023). A centralized clearing system, high retail participation, and significant foreign institutional investments portray India's unique market structure. This shift has extensive implications for domestic stakeholders and global markets monitoring these developments (Shah et al., 2008).

In the past two decades, settlement cycles worldwide have slowly shortened to improve market efficiency. The US Securities and Exchange Commission SEC authorized a T +1 settlement cycle effective May 2024, following concerns about margin requirements and liquidity constraints observed during the market volatility 2021 (Allen, 2024). Similarly, the European Union has continued with T+ 2 settlements, due to operational complexities in a multi-country financial system. While these global shifts focus to optimize capital efficiency, India's move to T+ 0 is a revolutionary step, setting it apart as an innovator in market infrastructure.

India's shift to same-day settlement is expected to have significant global implications, particularly for emerging markets exploring similar reforms (Yartey & Adjasi, 2007). A shorter settlement cycle minimizes counterparty risks and reduces the systemic impact of financial shocks. However, challenges persist, and Foreign Institutional Investors (FIIs), in particular, face hurdles due to prefunding requirements, currency exposure, and operational constraints(Bheemaiah & Bheemaiah, 2017). This study critically examines how these factors influence market behaviour and liquidity dynamics in the context of India's financial ecosystem.

This study employs an event study methodology to evaluate the impact of the T+ 0 settlement cycle on stock returns, trading volumes, and market liquidity. By comparing pre- and post-implementation market behaviour, the study aims to provide empirical insights into investor reactions, price efficiency, and regulatory effectiveness. The findings will contribute to the ongoing discourse on market structural reforms and serve as a reference for global policymakers and financial institutions evaluating shorter settlement cycles. Leading the charge toward an shortened settlement framework, India is setting a precedent for market efficiency and flexibility, influencing regulatory shifts in other jurisdictions (McCormick, 2010).It serves as a contemporary study for financial markets worldwide, offering understandings into the benefits, challenges, and broader implications of reducing settlement cycles.

The paper is organized as follows. The remainder of the paper is as follows: Section II discusses the literature review. Section III provides details about the Research Methodology. Section IV explains Data interpretations and Results. Section V provides Discussions and Implications. Section VI presents the Conclusion.

2 Theoretical Framework and Hypotheses Development

As per the semi-strong form of the Efficient Market Hypothesis (Fama, 1970), stock prices quickly and precisely replicate all publicly accessible material, including the implementation of the beta version of the T+ 0 settlement cycle in the Indian equity

markets. This implies that as the new settlement system is introduced, the market quickly adjusts to reflect its anticipated impact on liquidity, efficiency, and transaction costs (Fama, 1970).

The settlement cycle plays an important role in the systematic execution of trades in the securities markets (Schwartz & Francioni, 2004). It implies the duration between the trading day and the final settlement of the trade transaction, where the purchaser gets the securities and the seller receives the payment. Exchanges in different regions and countries follow settlement cycles based on their infrastructural abilities, with T + 2 settlement commonly followed (Mills et al., 2016).

2.1 *T + 2 Settlement cycle*

Figure 2 presents the T + 2 settlement cycle. It shows that on the Trade Day, when traders or investors put orders, they are executed after brokers confirm them, leading to obligations for both the buyer and seller. On the T + 1 business day, the custodians verify the trade details with the clients, calculate the obligations per the market prices, and further inform the clearing corporation (Loader, 2019). On the second business day, the buyer pays for the securities, and the seller delivers the securities. The securities and funds are exchanged between the buyer and seller, completing the settlement process. Previous studies have shown that implementing the T + 2 settlement cycle has increased market efficiency. For instance, the impact of implementing the T + 2 settlement system in the Saudi financial market was examined, revealing increased market efficiency and the facilitation of trading financial derivatives such as futures and options (Ramady, 2021). Additionally,

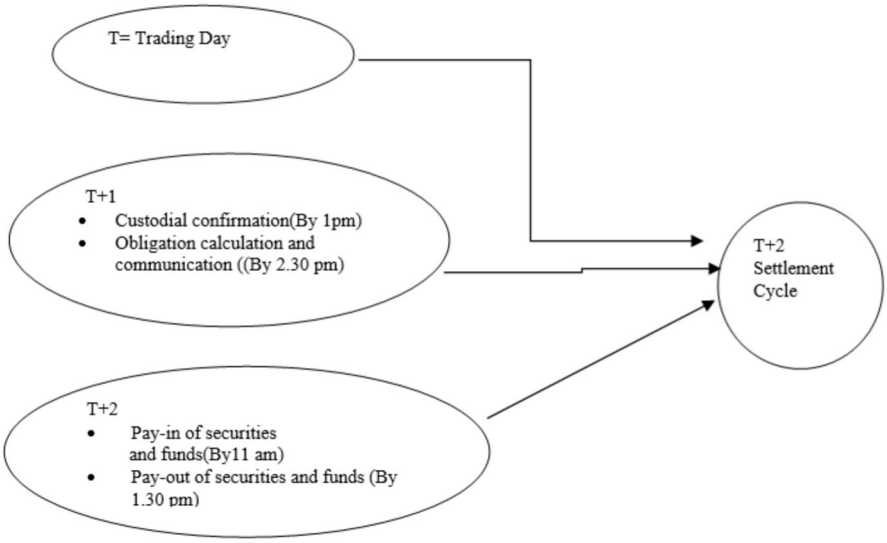

Fig. 2 T+2 Settlement Cycle. Source: Sebi.gov.in

the benefits of short selling for speculation and risk management were highlighted, with recommendations for adherence to international standards for market advancement (Abobaker, 2023).

2.2 *T + 1 Settlement*

Figure 3 presents the T + 1 settlement process, in which the custodial confirmation, obligation calculation and communication, and Pay-in and pay-out of securities and funds are executed on the next business day after the trading day.

2.3 *Migration from T + 3/T + 2/T + 1 to T + 0 settlement cycle*

Implementing the T + 2 from the T + 3 settlement cycle in the US helped improve liquidity for US-listed stocks compared to London-listed stocks, proving that reduced settlement helps increase market liquidity. It was also observed that the underlying securities that were difficult-to-borrow witnessed the most profound impact in terms of liquidity (Baig et al., 2022). In another study, the impact of different settlement cycles on the Indian stock market's efficiency was studied, focusing on calendar variances, especially the Day of the Week effect. The variables studied were unusual patterns of daily stock prices and stock returns. The study indicated that, before the introduction of T + 2 settlement, negative returns were observed on Monday and Thursday. In contrast, positive returns were found on all days after implementing the T + 2 settlement with the help of the GARCH MODEL (Patel & Mallikarjun, 2014). Another paper compares fixed weekly settlements with rolling T + 5, T + 3, and T + 2 periods. A dummy regression model is used to evaluate the presence of the Day of the Week effect in different settlement regimes. A significant impact on returns was observed during the fixed-day system but got reduced with rolling settlement. Also, no day-of-the-week effect was observed post the introduction of the rolling settlement system (Badhani, 2007).

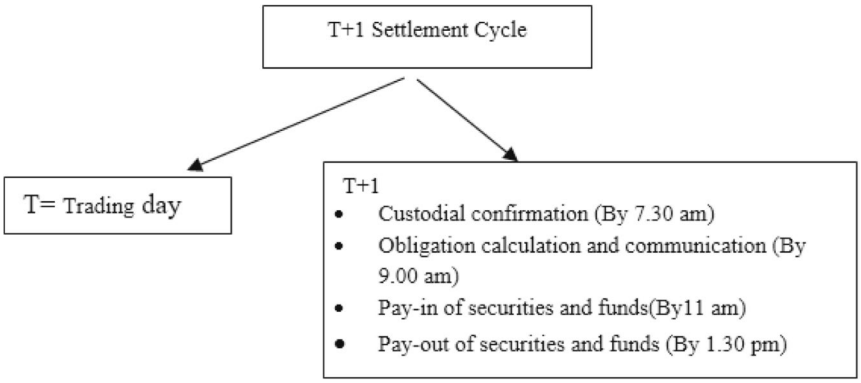

Fig. 3 T+1 Settlement Cycle. Source: Sebi.gov.in

The study explores the shift from T + 2 to T + 1 settlement cycle in global markets, driven by regulatory bodies. It highlights the impact on market participants and provides recommendations for the readiness for this transition (Cassells, 2023). The study highlights that reducing the US securities settlement cycle from T + 2 to T + 1 helped minimize market risks and costs. However, the transition is complex due to the associated nature of global financial markets, encompassing areas like derivatives, securities lending, and foreign exchange. (Abel, 2022). The study aimed to measure the impact of shortening the settlement cycle in equity markets from T + 2 to T + 1 on market liquidity, especially for less liquid stocks. Difference-in-Differences techniques was used to compare securities transitioning to T + 1 (treatment group) with those remaining at T + 2 (control group). The main variables analysed were the settlement cycle as the independent variable and market liquidity, measured by impact cost, as the dependent variable. The findings revealed a significant reduction in impact costs, indicating improved liquidity post-settlement cycle reduction. This positive effect was especially pronounced for smaller market capitalization stocks, suggesting that shortening settlement cycles can enhance overall market liquidity, with notable benefits for less liquid securities (Bhanu et al., 2024).

One of the studies investigated whether anomalous patterns in stock returns on any particular day of the week, were consistent after introducing compulsory rolling settlement on Indian stock exchanges. It was found that during the pre-rolling settlement period, Friday returns were the lowest, indicating the presence of the weekend effect. But, in the post-rolling settlement period, Friday returns were the highest, and Monday returns were the lowest (Chander & Mehta, 2007).

The impact of increased trading hours was studied on the volatility of the RUB/USD exchange rate in the Moscow Exchange Market, covering both same-day and next-day settlements from 2005 to 2013. Trading hours (as the independent variable), volatility (measured through historical and realized volatility), volume, and spreads (as dependent variables) were the key variables. Tick-by-tick transaction-level data was used to conduct the analysis at 5- and 15-min intraday intervals, both pre and post-the extensions. The researchers applied an ARMA-GARCH model to evaluate realized volatility, and a trade classification rule was used to estimate the probability of informed trading. The results indicated that volatility and volume significantly increased by extending trading hours, with volatility spikes being most pronounced after the market opening. A strong positive effect on realised volatility was found to be had by the length of the trading hours extension. Notably, informed trading increased substantially after the market opening for same-day settlements, whereas no such increase was observed for next-day settlements. While more opportunities for transactions and liquidity were provided by trading hours extensions, increased market volatility was also observed at the opening and midday sessions, suggesting that informed traders gained an edge over liquidity providers by the extensions (Frömmel & Kadioglu, 2023).

The impact of settlement rule changes made by the US National Securities Clearing Corporation on settlement efficiency was measured by failed share deliveries. The researchers hypothesised that these changes would impact both failed deliveries and market returns, and studied their bidirectional relationship

using a Vector Autoregressive model and Granger causality tests. A bidirectional relation between failed deliveries and market returns was observed. The expanded VAR model also showed that changed rules significantly affected the quantity of failed deliveries and market returns. These findings suggest that settlement process modifications can lead to the changes which may impact market dynamics and efficiency providing important observations for securities market design and trading strategies at the individual level (Asmar & Trimbath, 2022; The Global Treasurer, 2024).

The benefits, challenges, and best practices for a smooth transition to the T+ 1 settlement cycle was compiled by emphasizing the importance of technology in streamlining operations, managing risks, and increasing efficiency. With advanced technology and continuously improved approach, financial institutions can meet regulatory requirements and improve overall performance and customer experience (Ryan et al., 2023).

In October 2024, the Hong Kong Exchange confirmed its partnership with Digital Asset to develop a blockchain platform to allow global asset managers to complete post-trade allocations and processing for Stock Connect almost instantaneously. All market participants and counterparties will have real-time visibility of the post-trade status, allowing immediate actions to be undertaken. This can mitigate the risk of buy-ins while complying with the no-fail T+ 0 trade settlement requirement for China A-shares, which only allows four hours of operation on post-trade confirmation and allocation after market close (Tay, 2018).

The study analysed stock returns listed on Kuala Lumpur Stock Exchange from 1983 to 1993, finding negative returns on Mondays and favourable returns on Wednesdays and Thursdays. These patterns, unexplained by typical factors, suggested a market anomaly. Using an ARCH-M model, the positive returns were linked to higher market risk, but Monday's negative returns remained unexplained. After a settlement system change in 1990, these seasonal patterns largely disappeared, highlighting the influence of market structure on anomalies (Clare et al., 1998).

The study aimed to assess the impact of changes in the settlement regime on day-of-the-week returns in the New Zealand Stock Exchange from 1989 to 1996. Settlement practices and day-of-the-week stock returns were identified as the key variables. A time-series approach and orthogonal contrasts were used in the methodology, which was focused on isolating the effects of settlement changes without the confounding influences typically found in cross-market analyses. The results showed limited support for traditional settlement-based explanations of day-of-the-week returns. However, a pattern of depressed Monday returns was revealed, consistent with findings in other markets (Keef & McGuinness, 2001).

The objective of securities market settlement is to have real-time processing on a gross basis. But, these systems experience a form of delayed real-time processing, which takes place several days after a trade is made. Typically, securities settlements are processed on a T +3 basis, although T+ 5 systems can also be found. Because securities settlement systems involve significant risks, the current trend is to shift from deferred net settlement to deferred real-time settlement. Internationally, the goal has been to move toward T+ 1 real-time settlement, though this has not yet been fully achieved. Reducing risks, speeding up settlement, and removing

obstacles for efficient cross-border transfers have been important issues in securities settlement. Eventually, a securities settlement should be done with real-time T+ 0 processing, where settlement occurs immediately after the deal is made(Leinonen & Soramaki, 2011).

No study presents the migration from the T+ 1 to the T+ 0 settlement cycle. Hence, the following hypothesis has been made:

- H1: T+ 0 settlement did not significantly impact the cumulative abnormal returns and cumulative average abnormal returns of sample stocks.
- H2:There was no significant impact of post-T +0 settlement on trading volumes.

2.4 Comparison of Settlement Practices in Developed and Emerging Markets

Settlement cycles vary significantly between developed and emerging markets, influenced by factors such as market infrastructure, regulatory frameworks, and technological capabilities. Developed economies, such as the US and Europe, have systematically shortened settlement cycles over time, moving from T+ 5 to T+ 3, T+ 2, and most recently, T+ 1 in the US (Abel, 2022). The transition has been gradual, with regulators focusing on balancing risk reduction with market stability. These economies benefit from vigorous clearing and settlement infrastructure, high levels of institutional participation, and developed capital markets (Cassells, 2023). Emerging markets, on the other hand, face unique challenges. While India's shift to T+ 0 is unprecedented, other markets, like Brazil, South Korea, and South Africa, continue to operate on a T +2 cycle due to liquidity constraints and foreign investor concerns (Stanley, 2017). Many emerging markets have more retail investors than developed economies, which creates different market trends. Research shows that shorter settlement cycles can increase market swings in emerging markets, especially for less-traded stocks, as seen in Saudi Arabia after moving to T+ 2 (Jenkins et al., 2019).

A significant distinction between developed and emerging markets is prefunding. While US and European institutions can access sophisticated clearing mechanisms that minimize funding requirements, Indian markets require higher collateral deposits, potentially affecting FIIs. These differences suggest that the impact of settlement cycle reductions is not unchanging across markets, necessitating region-specific regulatory considerations (Sharma et al., 2024).

2.5 *Role of Technological Advancements in Facilitating T+ 0 Settlements*

Technological innovations such as blockchain, distributed ledger technology (DLT), and AI have significantly enhanced the feasibility of T+ 0 settlement. Traditional clearing and settlement systems depend on centralized databases and intermediaries, leading to delays in reconciliation and increased counterparty risk. Introducing blockchain-based RTGS models has eliminated manual reconciliation, reducing trade failures and accelerating settlement times (Macharia, 2023). In India, the feasibility of blockchain-based clearing mechanisms is under discussion, with

financial institutions exploring ways to reduce the dependence on prefunded margins while ensuring trade security. However, adoption challenges remain concerning interoperability with legacy systems, regulatory approvals, and cybersecurity risks (Priem, 2020). Despite these challenges, research suggests that DLT can streamline settlement processes by reducing intermediaries, thereby supporting India's T + 0 ambitions

2.6 Implications of Shortened Settlement Cycles on Market Liquidity and Investor Confidence.

Existing literature indicates that reducing settlement cycles increases liquidity and introduces market complexities. A U.S.-based study found that transitioning from T + 2 to T + 1 reduced impact costs and improved market depth, particularly for small-cap and less liquid stocks (Swartley et al., 2010). Similar effects were observed when the US moved from T + 3 to T + 2, with increased liquidity in previously hard-to-borrow securities (Tang, 2024).

However, for emerging markets, shortened settlement cycles may lead to capital restraints for foreign investors, who require supplementary time for currency conversion, compliance checks, and funding transactions. The shift from T + 3 to T + 2 in India transformed intraday price movements, with studies showing a disappearance of negative Monday returns post-implementation, gesturing increased efficiency (Jaisinghani, 2016). Investor confidence also plays an important role in settlement cycle reforms. Research on New Zealand's stock market found that changes in settlement practices influenced day-of-the-week return anomalies, suggesting a behavioural shift among investors (Symons, 2018). Despite these findings, to our knowledge, no inclusive study is available on the effects of transitioning from T + 1 to T + 0. This research gap emphasizes the need for empirical analysis of how immediate settlement cycles influence market efficiency, liquidity, and investor confidence, especially in retail-driven emerging markets such as India. Integrating blockchain and AI-driven clearing mechanisms may further redefine post-trade processing, ensuring market efficiency and investor confidence are maintained in an accelerated settlement framework. However, empirical research on T + 0 remains limited, necessitating further analysis of its real-world impact (Lin & Chen, 2019) (Bhanu et al., 2024).

3 Research Methodology

The section discusses the research methodology used to study the impact of T + 0 settlement.

3.1 Sample Stocks

The authors selected all 25 stocks listed by the national stock exchange from different sectors for the T + 0 settlement pilot launch (Table 1).

Table 1 List of T + 0 Companies and their Industries

Name of Company	Industry
Ambuja Cements	Cement
Ashok Leyland	Automotive
Bajaj Auto	Automotive
Bank of Baroda	Banking
BPCL	Oil & Gas
Birla soft	Information Technology
Cipla	Pharmaceuticals
Coforge	Information Technology
Divi's Laboratories	Pharmaceuticals
Hindalco Industries	Metals & Mining
IHC	Hospitality
JSW Steel	Metals & Mining
LIC Housing Finance	Financial Services
LTIMindtree	Information Technology
Motherson International	Automotive Components
MRF	Tires
Nestle India	Food & Beverages
NMDC	Metals & Mining
ONGC	Oil & Gas
Petronet LNG	Oil & Gas
SBI	Banking
Tata Communications	Telecommunications
Trent	Retail
Union Bank of India	Banking
Vedanta	Metals & Mining

Source: National Stock Exchange

3.2 Data Collection

The data was collected from the National Stock Exchange of India from June 23, 2023, to April 30, 2024.

3.3 Variables

The study variables include CAR, CAAR and daily volume average of the sample stocks. For the proxy of a market index, Nifty Index has been used as a traditional market model to ascertain abnormal returns. Table 2 provides the details of the study's variables with brief explanation. Log returns are calculated for the closing stock price returns with the help of the continuous compounding formula LN (T/T-1) and converted to percentages. T refers to trade Day and T-1 refers to Day 1.

Table 2 Summary of the variables, their codes, and definition

Variable	Code	Definition
CAR over the event window (−2 to 2)	CAR(−2,2)	Cumulative abnormal return calculated between 2 days before and 2 days after election results.
CAR over the event window (−5 to 5)	CAR(−5,5)	Cumulative abnormal return calculated between 5 days before and 5 days after election results.
CAR over the event window (−10 to 10)	CAR(−10,10)	Cumulative abnormal return calculated between 10 days before and 10 days after election results.
CAR over the event window (−20 to 20)	CAR(−20,20)	Cumulative abnormal return calculated between 20 days before and 20 days after election results.
CAAR over the event window (−2 to 2)	CAAR(−2,2)	Cumulative average abnormal return calculated between 2 days before and 2 days after election results.
CAAR over the event window (−5 to 5)	CAAR(−5,5)	Cumulative average abnormal return calculated between 5 days before and 5 days after election results.
CAAR over the event window (−10 to 10)	CAAR(−10,10)	Cumulative average abnormal return calculated between 10 days before and 10 days after election results.
CAAR over the event window (−20 to 20)	CAAR(−20,20)	Cumulative average abnormal return calculated between 20 days before and 20 days after election results.
Percentage Change in Average Volumes after T+0 introduction	%ΔVol(T+0)	Percentage change in average trading volumes (measured in number of contracts) after the introduction of T+0 settlement.

Source: Author's own work

To check stationarity in the variables, Augmented Dickey-Fuller (ADF) is applied tests help evaluate the existence of unit roots in log return time series, ensuring the validity of observations in event studies(Chang et al., 2018).

Robustness Check: Fixed Effect (FE) Model, Hausman Test, Crude Dependence Adjustment (CDA) and Generalized Sign (G Sign Z) tests were applied to enhance the credibility of the results.

The fixed-effects model aims to control unobserved and time-invariant features that may be linked with the independent variables in the model. To validate the application of the FE model, the rejection of the null hypothesis in the Hausman test, implying that the FE model is preferred over the Random Effects model, indicates significant cross-sectional heterogeneity (Lebo & Weber, 2015).

CDA is a parametric test that adjusts for cross-sectional dependence using a single, sample-wide variance estimate. A high CDA shows significant abnormal

returns. Low CDA values, however, show that the average abnormal returns during the event window are not statistically significant after considering cross-sectional dependence. This suggests that the magnitude of the average abnormal returns is not large enough to be considered a real effect of the event.

G Sign Z, on the other hand, is a non-parametric test linking positive abnormal return proportions. The numbers show that the event studied had very little effect on stock returns. The values are close to zero, meaning no significant changes occurred. Low G Sign Z values, however, show that the proportion of positive abnormal returns during the event window is not significantly different from what would be expected during the normal estimation period.

This suggests that the event did not have a consistent positive or negative directional impact on the returns of the sample stocks (Rani et al., 2013) (Campbell et al., 2010).

3.4 Statistical Technique

Event Study Methodology and Dynamic Forecasting Technique were used to test the hypothesis for the study's objectives statistically.

The statistical technique used for the objectives was event study methodology with 28–03–2024 as the event date with event windows of 2, 5, 10, and 20 days before and after the event (Table 2) (Jegarut et al., 2021; Dzar et al., 2023; Pinyatanabat & Kluaymai-ngarm, 2024; Basha et al., 2021; Hasan & Al-Najjar, 2024).

Justification for Event Windows (2, 5, 10, and 20 days). The event windows are designed to study both immediate and continued market reactions. 2-day window studies immediate market reactions, typically driven by algorithmic and high-frequency traders (Ghoshal, 2018).

Five-day windows allow for short-term adjustments, reflecting how institutional investors adjust to the news. A 10-day window offers an understanding of medium-term market behaviour due to liquidity shifts and fund rebalancing. A 20-day window analyses the longer-term impact, including investor sentiment shifts and potential volatility stabilization.

Dynamic forecasting is employed to analyse market trends post-T + 0 settlement adoption. This technique is essential in the Indian market because of high retail investor participation. Retail investor sentiment heavily influences India's market response to new settlement systems, which differ from developed markets and require a flexible predictive approach. The introduction of T+ 0 settlement adjusts liquidity cycles and intraday volatility patterns, which traditional forecasting models may not fully reflect (Zhang, 2017). Furthermore, since SEBI continuously hones settlement practices, dynamic forecasting helps calculate the long-term market course beyond the immediate event window.

Also, an estimation window of 167 trading days before the event window was taken considering from day –167 to day –1, where day 0 is when the beta version of T+ 0 settlement was introduced in the case of 25 stocks. The estimation window calculates the expected returns for the event study. One of the existing studies considered an estimation window of 120 days. An estimation window of more than

120 days is deemed reliable to the present work (Ledwani et al., 2021). The abnormal returns are calculated by integrating the market model, calculating the expected returns constructed on the association between stock returns and market returns in the estimation window. CAR is calculated for each stock, and CAAR are computed for the sample period in the 2,5,10, and 20-day event windows. Brown and Warner's Model of Event Study has been used. Event studies help in underscoring how underlying securities respond to a given event (Brown & Warner, 1985).

Statistical significance is found using t-tests for individual stock CARs and sample period CAARs to assess the overall impact during the study period. A volume analysis based on percentage changes in trading volume post T + 0 settlement introduction is also undertaken. The research design guides analyse how the introduction of T + 0 settlement impacts stock returns and market dynamics in the Indian equity market. This will help design the policy measures and their implications for market participants and other stakeholders.

Dynamic forecasting was applied to compare the forecast values with the actual closing price values during the entire 41-day event window. The accuracy of the comparison was then analysed using the statistical metrics such as Root Mean Square Error (RMSE), Mean Absolute Error (MAE), Mean Absolute Percent Error (MAPE), Theil inequality coefficient (TIC), and Bias Proportion, along with p values and coefficient.

RMSE calculates the square root of the average squared gaps between forecast and actual values. A lower RMSE value points towards higher model accuracy. MAE analyses the average of complete gaps between forecast and actual values. MAPE calculates the average absolute per cent gap between forecast and actual values, showing accuracy as a percentage. TIC associates the precision of a predictive model with a random walk model ranging from 0 to 1, where values nearer to 0 show a better relation between forecast and actual values. Bias Proportion shows whether the model over-predicts or under-predicts. A higher value confirms that the model diverges from the actual values in one direction, highlighting model omission.

For dynamic forecasting, the variables"closing prices"from 23–06–2023 to 26–02–2024 were used as the estimation period, whereas 27–02–2024 to 30–04–2024 was treated as the forecast period for each stock. Following the dynamic regression equation (with lag values of the dependent variable as the independent variable) for univariate series was applied:

$$y \; c \; y(-1)$$

where y is dependent variable and y(−1) is the independent variable.

4 Data Interpretations and Results

The results and data interpretation of the research methodology have been mentioned in this section: Table 3.

Table 3 and 4 above provides the descriptive statistics for the individual equities of Previous Studies and daily trade-to-trade returns 25 Sample Stocks respectively.

Table 3 Descriptive statistics of daily trade-to-trade returns individual equities of Previous Studies

Country	Number of Stocks	Mean no. of returns per stock	% of the overall sample	Mean across stocks of:					
				Mean	Median	Standard deviation	Skewness	Excess Kurtosis	Percent of zero returns
Overall	48258	1665	100.00%	0.008	0.001	2.696	4.811	229.832	20.7%
Argentina	135	1350	0.20%	0.171	0.131	2.847	3.015	88.436	14.2%
Australia	2263	1743	3.10%	0.009	0.007	2.713	2.978	84.015	12.6%
Austria	228	1646	0.50%	0.006	0.005	0.073	5.961	164.610	13.9%
Belgium	886	1718	1.30%	0.006	0.003	4.866	3.745	146.819	13.2%
Brazil	798	2036	1.30%	0.122	0.027	3.744	8.645	111.260	21.8%
Canada	678	1614	1.30%	0.016	0.007	2.931	7.888	58.127	14.5%
China	1435	1897	3.00%	0.000	0.000	2.030	9.564	58.157	13.2%
France	2094	1542	3.40%	0.012	0.004	0.531	3.916	176.802	12.6%
Germany	6306	1912	10.40%	0.004	0.002	0.545	4.031	152.940	12.1%
India	1315	1966	3.20%	0.004	0.010	0.079	1.813	67.554	7.70%
Italy	565	2436	1.70%	0.464	0.192	17.352	20.866	731.872	15.1%
Japan	3715	2663	12.30%	0.382	0.025	19.914	8.889	511.614	12.4%
UK	5398	1847	12.60%	0.012	0.003	3.678	6.207	314.048	14.4%
Zimbabwe	2	444	0.00%	0.029	0.029	2.110	5.029	139.258	28.1%

Source: (Campbell et al. 2010)

Table 4 Descriptive statistics of daily trade-to-trade returns 25 Sample Stocks

Mean no. of returns per stock	% of the overall sample	Mean across stocks of:						
		Mean	Median	Standard deviation	Skewness	Excess Kurtosis	Percent of zero returns	Percent of zero returns
25	208	0.005	0.083	0.0297333	4.4072667	6.4905333	202.35627	7.70%

Source: Author's own work

4.1 ADF Unit Root Test: the Test was Applied to Check Whether the Variables were Stationary

All the variables were stationary at the first difference using the ADF Unit Root Test as evident from Table 5:

4.2 Fixed-effects Model Results

Table 6 shows the results of the fixed-effect model. The R^2 values between 0.552 and 0.612 show a rational fit, implying that NIFTY(a proxy for the market index) explains around 55%–61% of the variation in CAR.

F-test results are also statistically significant, with a p-value of less than 0.01, which validates the overall strength of the model(Kumar & Arora, 2012). Since

Table 5 Stationarity Results for closing prices of 25 stocks using ADF Unit Root Test

Stock Name	Augmented Dickey-Fuller Test	
	t-Stat level	Prob value
Ambuja Cements	–13.3428	0.0
Ashok Leyland	–14.8518	0.0
Bajaj Auto	–14.6028	0.0
Bank of Baroda	–13.9131	0.0
BPCL	–13.5593	0.0
Birla soft	–15.3229	0.0
Cipla	–14.1983	0.0
Coforge	–12.8278	0.0
Divi's Laboratories	–14.1072	0.0
Hindalco Industries	–14.0240	0.0
IHC	–15.9342	0.0
JSW Steel	–15.6377	0.0
LIC Housing Finance	–13.7470	0.0
LTIMindtree	–14.7408	0.0
Motherson International	–15.3003	0.0
MRF	–14.6847	0.0
Nestle India	–17.7693	0.0
NMDC	–12.9944	0.0
ONGC	–15.4603	0.0
Petronet LNG	–12.8086	0.0
SBI	–13.4241	0.0
Tata Communications	–14.6417	0.0
Trent	–11.8773	0.0
Union Bank of India	–14.1938	0.0
Vedanta	–12.8194	0.0

Source: Author's computation/compilation (obtained from EViews)

the Fixed Effect model is applied, it indicates that firm-specific differences impact CAR over time (Table 6).

Justification for applying FE Model: Table 7 provides the values for justifying the adoption of a FE model after applying Hausman Test.

- H_0: Cross Sectional heterogeneity is insignificant

As per Table 7, a p-value of 0.034 is less than 0.05, rejecting the above null hypothesis, inferring that cross-sectional heterogeneity is significant and that the FE model is the most appropriate.

4.3 Event Methodology Results

The results of the event study methodology in the event windows of 2, 5, 10, and 20 days before and after the introduction of T+0 settlement have been mentioned in this section. T+0 settlement launch in India for 25 stocks has shown a mixed response. Table 8 presents the beta coefficients of cumulative abnormal returns (CARs) at the 5% significance level, along with the corresponding t-values and robustness coefficients for the sample stocks, as represented by their G-signal values and CDA values.

There was a significant increase in the returns of Indian Hotels and Tata Communications, with a 6% abnormal return and a t-value of 2.21. Hindalco also recorded significant abnormal returns in 20-days window. Vedanta was the robust one with maximum positive impact. It is showing the consistent abnormal returns in all the covered event windows. There were abnormal returns of 8% for 2 days, 14% for 5 days, 31% for 10 days, and a substantial 36% for the 20-day event window. Interestingly, Hindalco also showed a positive impact, but only in the longer 20-day window, with an abnormal return of 18%. However, not all companies benefited. Coforge experienced a significant negative impact in the 10-day window, with abnormal returns of −14% and −13% respectively. Considering the 20-day window, the picture becomes even more varied. While Vedanta continued its positive trend, Birla Soft, Coforge, and MRF all suffered significant negative abnormal returns, ranging from −3% to a substantial −27% for Coforge. Therefore, the hypothesis that the introduction of T+ 0's beta version did not significantly impact the cumulative abnormal returns was accepted except in case of Vedanta, IHC, Tata Communications, Coforge, Birla Soft, MRF and Hindalco.

To validate the above findings, the CDA robustness coefficient values, which are found to be very close to zero and in some cases slightly negative, suggest that after accounting for cross-sectional dependence, no statistically significant average abnormal return is observed in any of the event windows overall. The small magnitude of the CDA values implies that implementation of beta version of T+ 0 settlement did not have a major impact on the returns of the study's variable overall.

In the shorter event windows (−2,2) and (−5,5), the G Sign Z robustness coefficients values are positive (0.84 and 0.45, respectively). These values are relatively

Table 6 Fixed Effect Model Results

CAR Event Window	Constant (β_0)	NIFTY (β_1)	R^2	F-test	*p*-value (*F*-test)
CAR(−2,2)	0.028	0.007	0.612	19.26	0
CAR(−5,5)	0.039	0.006	0.589	17.89	0
CAR(−10,10)	0.045	0.005	0.57	16.45	0
CAR(−20,20)	0.052	0.004	0.552	15.21	0

Source: Author's computation/compilation (obtained from EViews)

Table 7 Hausman Test Results

Test Statistic	Chi-Square Value	*p*-value	Result
Hausman Test	8.92	0.034	FE is preferred (H_0 Rejected)

Source: Author's computation/compilation (obtained from Stata

Table 8 CAR-Coefficient-Values and Robustness Coefficients for Sample Stocks

Event Window	CAR (−2,2)	*t*-value (−2,2)	CAR (−5,5)	*t*-value (−5,5)	CAR (−10,10)	*t*-value (−10,10)	CAR (−20,20)	*t*-value (−20,20)	G Sign Value$_z$	CDA Value$_t$
Ambuja Cements	0.05	1.83	0.03	0.74	0.02	0.32	–0.04	–0.54	0.34	0.59
Ashok Leyland	0.05	1.66	0.06	1.52	0.06	1.00	0.10	1.30	0.20	0.63
Bajaj Auto	–0.01	–0.41	–0.01	–0.15	0.00	0.02	–0.07	–0.83	0.24	0.38
Bank of Baroda	0.02	0.73	0.03	0.75	–0.04	–0.74	0.00	0.00	0.21	0.48
BPCL	0.01	0.38	0.00	0.03	–0.08	–1.48	–0.12	–1.46	0.11	0.44
Birla Soft	0.02	0.77	–0.02	–0.47	–0.12	–2.10	–0.30	–3.79	7.24	6.74
Cipla	0.00	–0.16	–0.01	–0.27	–0.08	–1.46	–0.10	–1.28	0.75	0.75
Coforge	–0.01	–0.43	0.01	0.27	–0.14	–2.49	–0.27	–3.38	7.00	6.91
Divi's Laboratories	0.04	1.60	0.06	1.57	0.05	0.84	0.11	1.40	0.80	0.84
Hindalco Industries	0.02	0.68	0.02	0.48	0.13	2.33	0.18	2.31	6.72	6.64
IHC	0.06	2.21	0.06	1.56	0.02	0.44	–0.08	–1.05	0.30	0.41
JSW Steel	0.05	1.66	0.03	0.79	0.07	1.16	0.08	1.02	0.90	0.64
LIC Housing Finance	0.04	1.55	0.06	1.58	–0.01	–0.11	–0.07	–0.87	0.58	0.48
LTIMindtree	–0.03	–1.12	–0.05	–1.30	–0.05	–0.91	–0.12	–1.56	0.22	0.50
Motherson International	0.02	0.69	0.03	0.75	0.02	0.33	0.06	0.78	0.51	0.10
MRF	0.03	1.16	0.00	–0.05	–0.11	–1.99	–0.16	–1.97	6.01	7.47
Nestle India	0.01	0.38	0.00	–0.01	–0.01	–0.23	–0.04	–0.54	0.62	0.21
NMDC	0.02	0.73	0.01	0.20	0.02	0.40	–0.06	–0.79	0.55	0.22
ONGC	0.01	0.21	–0.03	–0.62	0.00	0.04	–0.05	–0.63	0.32	0.17

Table 8 (continued)

Event Window	CAR (−2,2)	*t*-value (−2,2)	CAR (−5,5)	*t*-value (−5,5)	CAR (−10,10)	*t*-value (−10,10)	CAR (−20,20)	*t*-value (−20,20)	G Sign Value$_z$	CDA Value$_t$
Petronet LNG	0.03	1.22	0.03	0.62	0.09	1.57	0.03	0.43	0.54	0.60
SBI	0.01	0.19	0.01	0.33	–0.01	–0.24	0.03	0.44	0.86	0.47
Tata Communications	0.06	2.21	0.05	1.17	–0.03	–0.51	–0.08	–1.06	5.50	5.11
Trent	–0.04	–1.56	–07	–1.62	–0.09	–1.54	–0.03	–0.38	0.27	0.29
Union Bank of India	0.02	0.85	0.00	0.06	–0.09	–1.53	–0.09	–1.13	0.13	0.35
Vedanta	0.08	3.08	0.14	3.37	0.31	5.48	0.36	4.59	5.55	7.10

Source: Author's compilation

low, suggesting that the proportion of positive abnormal returns is somewhat higher but not statistically significant in shorter windows.

In the longer event windows (−10,10) and (−20,20), the G Sign Z values are negative (−0.06 and −0.36, respectively), indicating a slight tendency for less positive abnormal returns. However, the values are close to zero, signifying no significant impact.

Based on the above robustness coefficient values, it can be determined that implementing the beta version of T+ 0 settlement did not have a statistically significant impact on the abnormal returns of the sample stocks.

The results are consistent across all event windows, strengthening the overall coefficients. The analysis of 25 sample stocks shows mixed results, with 7 showing significant abnormal returns and 18 insignificant abnormal returns. Therefore, the implementation of T+ 0 settlement in beta version replicates semi-strong market efficiency by quickly incorporating public information (Polyzos & Wang, 2022).

Similarly, cumulative average abnormal returns were calculated for the entire sample of 25 stocks to analyse whether there was a significant average impact of T+ 0 settlement on the returns. It is evident from Table 9 that the CAAR for the window spanning from two days before and two days after the T+ 0 settlement is 0.02, indicating a 2% increase in average abnormal returns. The corresponding t-value is 0.84. Since a t-value greater than 1.96 (95% confidence level) or even more significant than 1.64 (90% confidence level) would indicate statistical significance, a t-value of 0.84 suggests that the observed CAAR is not statistically significant. This implies that the returns in this short-term window are not significantly different from zero. The CAAR for the 5-day window within the T+ 0 settlement is also 0.02, or 2%. The t-value for this window is 0.45, much lower than the threshold needed for statistical significance. This suggests no significant impact on returns in the medium-term window around the T+ 0 settlement. The CAAR is 0.00 in the 10-day event windows, implying that no cumulative average abnormal return was observed during this period. The T-value of −0.06 also does not reflect any statistical significance. This means that the T+ 0 settlement did not pointedly impact returns in the 10-day event window. Similarly, in the 20-day event windows, the CAAR is found to be −0.03, displaying a nominal decrease in average abnormal return of 3%. The T-value of 0.36 also does not reflect any statistical significance, implying that the T+ 0 settlement did not pointedly impact returns in the 10-day event window.

Therefore, it is evident from Table 9 that CAARs for the 2,5,10,20 event windows for the event of the T+ 0 settlement in the case of the 25 sample stocks also did not

Table 9 CAAR-Coefficient-Values & Robustness Coefficients for T+0 Stocks

Event Windows	CAAR	*t*-value	CDA Value$_t$	G Sign Value$_z$
(−2,2)	0.02	0.84	0.95	0.28
(−5,5)	0.02	0.45	0.82	2.17
(−10,10)	0.00	–0.06	0.22	1.60
(−20,20)	–0.03	–0.36	1.25	1.47

Source: Author's compilation

show abnormal returns in a statistically significant manner. The t-values in each case are less than the required thresholds to prove the statistical significance, indicating that the T + 0 settlement did not have a notable impact on stock returns in the analysed periods. Therefore, the hypothesis was accepted that introducing T + 0's beta version did not significantly impact the cumulative average abnormal returns.

Based on the robustness coefficient values for CAAR also, it can be concluded that the implementation of the beta version of T + 0 settlement did not have a statistically significant impact on the CAAR of the sample stocks. The t-values for all event windows remain below significance thresholds, while the robustness checks using CDA and G Sign tests also indicate no firm evidence of abnormal performance around the event (Rani et al., 2013). The insignificance of cumulative average abnormal returns coincides with the weak form of market efficiency, highlighting that historical stock prices alone may not predict future values (Espinosa-Paredes et al., 2022).

In the Research Methodology section, an estimation window of 167 trading days before the event window was used, covering the period from day –167 to day –1. Day 0 represents introducing the beta version of T + 0 settlement for 25 stocks. To compare the average volumes during the estimation window (23–06–2023 to 26–02–2024) and the forecast window (27–02–2024 to 30–04–2024), the percentage change in average volumes after the T + 0 introduction was presented in Table 10.

The data provides information about the percentage change in average trading volumes for 25 stocks after introducing the T + 0 settlement beta version. These changes reflect a mixed response, with some stocks experiencing increases in average volumes and others seeing declines.

Sixteen out of the 25 stocks experienced an increase in average trading volumes. Among these 16 stocks, ten witnessed significant increases of more than 5%, indicating a strong positive market response to the T + 0 settlement change. These stocks are Coforge (+ 9.00%), Hindalco Industries (+ 8.00%), MRF (+ 6.00%), Nestle India (+ 11.00%), NMDC (+ 8.00%), ONGC (+ 11.00%), Petronet LNG (+ 18.00%), Trent (+ 13.00%), BPCL(+ 7%)and Vedanta (+ 18.00%). Notably, Petronet LNG and Vedanta registered the highest increase of 18.00% in average volumes, demonstrating a strong market response following the launch of the beta version of T + 0 settlement.

On the contrary, nine stocks saw a decrease in average trading volumes. Five experienced a decline of more than 5%, signifying a more negative market reaction. These stocks are Birlasoft (−9.00%), JSW Steel (−8.00%), LTIMindtree (−10.00%), Motherson International (−6.00%), and Tata Communications (−6.00%). LTIMindtree witnessed the most significant decline, with a −10.00% drop in average volumes.

Therefore, the H2 hypothesis that there was no significant impact of post-T +0 settlement on trading volumes was rejected except in case of Ambuja Cements, Bajaj Auto, Bank of Baroda, Cipla, Divi's Laboratory, IHC, LIC Housing Finance, SBI and Union Bank of India at 5% significance level. The introduction of the T + 0 settlement had a varied impact on trading volumes. While 15 stocks showed increases, particularly Petronet LNG and Vedanta with 18.00% increases, 10 stocks experienced declines, with LTIMindtree and Birlasoft seeing notable reductions.

Table 10 List of Percentage Change in Average Volumes after T+0 introduction

Stock Name	Percentage Change in Average Volumes after T+0 introduction
Ambuja Cements	–2.00%
Ashok Leyland	5.00%
Bajaj Auto	2.00%
Bank of Baroda	–1.00%
BPCL	7.00%
Birlasoft	–9.00%
Cipla	–4.00%
Coforge	9.00%
Divi's Laboratories	5.00%
Hindalco Industries	8.00%
IHC	1.00%
JSW Steel	–8.00%
LIC Housing Finance	2.00%
LTIMindtree	–10.00%
Motherson International	–6.00%
MRF	6.00%
Nestle India	11.00%
NMDC	8.00%
ONGC	11.00%
Petronet LNG	18.00%
SBI	2.00%
Tata Communications	–6.00%
Trent	13.00%
Union Bank of India	–1.00%
Vedanta	18.00%

Source: National Stock Exchange

The diverse outcomes underscore the dynamic nature of market reactions to policy changes such as introducing the beta version of T + 0 settlement.

The changes in trading volumes post the beta implementation of the T + 0 settlement mechanism may be due to several factors. Stocks witnessing increased volumes profited from increased liquidity with shortened settlement, mitigates counterparty risk and fascinates high-frequency traders. Furthermore, stocks with high retail participation, like energy, consumer goods, and metals, may have gained adhesion owing to increased speculative interest. Institutional investors may have also preferred certain stocks for inclusion in short-term portfolios, capitalizing on the fast flow of money due to T + 0 settlement.

On the other hand, stocks witnessing a decrease in volumes can be affected by the diligence exercised by institutional investors, since they prefer longer settlement cycles citing operational efficiency. IT and manufacturing stocks, which generally attract long-term investors compared to short-term traders, might have seen less

action due to the settlement cycle's inadequate impact on their long-term investment strategies. Similarly, stocks with lower volatility in the past may not have gained significant trading interest, resulting in lower volume changes.

In stocks where changes were less than a 5% change in volumes, the impact of T + 0 settlement may have been less affected owing to balanced investor sentiment. The attributed reasons may be liquidity issues, specifically in utilities or healthcare, or where investors believe in long-term investment compared to intraday trading or short-term investment.

Overall, the mixed responses of trading volumes indicate that while T + 0 settlement increases liquidity and market efficiency, its impact is more stock-specific, showcasing the semi-strong form of market efficiency (Cullen, 2023).

4.4 Dynamic Forecasting Technique

In this section, the data interpretation of the statistical techniques such as RMSE, MAE, MAPE, TIC and Bias Proportion values for each of the 25 stocks are analysed.

Table 11 shows the RMSE, MAE, MAPE, TIC and Bias Proportion values for each of the 25 stocks studied. The higher RMSE values in the case of Vedanta, Indian Hotels, Tata Communications, Hindalco, Coforge, Birla Soft, and MRF indicate a significant difference between the forecast and closing price values. RMSE is used to measure the accuracy of a predictive model; in the context of stock prices, it quantifies the difference between predicted and actual values. Lower RMSE values indicate better model accuracy, while higher values suggest less accurate predictions. Birla Soft has an RMSE of 94.9, Coforge has a much higher RMSE of 943.21, and Hindalco Industries shows a RMSE of 70.13. IHC has an RMSE of 130.07, while MRF exhibits a highly high RMSE of 26,567. Tata Communications has an RMSE of 151.21, and Vedanta has a RMSE of 76.95.

Likewise, TIC, closer to 0, shows a better relation between forecast and actual values, ranging from 0.04 to 0.13, further supporting our findings. The TIC values for the stocks are as follows: Birla Soft has a TIC of 0.06, Coforge has a TIC of 0.07, and Hindalco Industries also has a TIC of 0.06. IHC has a TIC of 0.1, MRF has a TIC of 0.08, Tata Communications has a TIC of 0.04, and Vedanta has a TIC of 0.13.

The bias Proportion is also greater than 0.51 in the above seven stocks, indicating a significant difference between the forecast and closing price value. The Bias Proportion values for the stocks are as follows: Birla Soft has a Bias Proportion of 0.65, indicating that 65% of the prediction error is due to systematic bias in the model. Coforge has a higher Bias Proportion of 0.77, meaning 77% of the error stems from bias. Hindalco Industries has a Bias Proportion of 0.64, suggesting that 64% of the error is due to bias. IHC has a Bias Proportion of 0.71, showing that 71% of the error comes from systematic bias. MRF has the highest Bias Proportion at 0.78, indicating significant bias in the model's predictions. Tata Communications has a Bias Proportion of 0.58, which is relatively lower but still

Table 11 Statistical Validity of Forecast Prices based on Estimated Prices using Dynamic Forecasting

Stock Name	Co-efficient	*P*-Value	RMSE	MAE	MAPE	TIC	Bias Proportion
Ambuja Cements	1.00	0.0	43.75	39.34	6.48	0.034	0.80
Ashok Leyland	0.91	0.0	7.54	5.85	3.45	0.02	0.29
Bajaj Auto	1.01	0.0	1143.81	960.84	10.92	0.06	0.70
Bank of Baroda	1.00	0.0	18.74	16.47	6.28	0.034	0.65
BPCL	1.01	0.0	76.13	66.23	22.18	0.11	0.75
Birla soft	0.99	0.0	94.90	77.88	11.08	0.06	0.65
Cipla	0.98	0.0	42.51	32.75	2.31	0.014	0.05
Coforge	0.98	0.0	943.21	832.27	15.20	0.07	0.77
Divi's Laboratories	0.91	0.0	158.55	131.11	3.63	0.02	0.05
Hindalco Industries	0.97	0.0	70.13	56.65	9.48	0.06	0.64
IHC	1.01	0.0	130.07	110.28	18.87	0.10	0.71
JSW Steel	0.94	0.0	47.47	38.54	4.47	0.02	0.49
LIC Housing Finance	1.00	0.0	88.16	81.74	13.07	0.06	0.86
LTIMindtree	0.96	0.0	539.54	491.67	10.07	0.05	0.83
Motherson International	0.98	0.0	8.56	6.87	5.55	0.03	0.50
MRF	1.00	0.0	26567	23582	17.76	0.08	0.78
Nestle India	0.98	0.0	41.99	32.85	1.30	0.00	0.08
NMDC	0.99	0.0	27.88	23.42	10.91	0.05	0.68
ONGC	1.00	0.0	23.69	21.42	7.90	0.04	0.76
Petronet LNG	0.99	0.0	18.16	14.46	5.33	0.03	0.41
SBI	1.01	0.0	76.56	67.27	8.83	0.04	0.75
Tata Communications	0.94	0.0	151.21	129.75	6.58	0.04	0.58
Trent	1.00	0.0	340.59	286.55	7.10	0.04	0.68
Union Bank of India	0.99	0.0	5.45	4.45	2.97	0.02	0.02
Vedanta	0.97	0.0	76.95	55.52	15.56	0.13	0.51

Source: Author's computation/compilation (obtained from EViews)

indicates notable bias. Vedanta has the lowest Bias Proportion at 0.51, suggesting that 51% of the error is due to bias, which is comparatively lower than the other stocks. Overall, the Bias Proportion values above 0.50 indicate significant variation between forecasted and actual values, supporting the findings of abnormal returns in these stocks.

Figures. 4, 5, 6, 7, 8, 9 and 10 show the graphical representation of dynamic forecasting in the case of seven stocks that witnessed cumulative abnormal returns. Therefore,"H_{01}: T+ 0 settlement did not significantly impact the cumulative abnormal returns and cumulative average abnormal returns of included stocks."was accepted. However, cumulative abnormal returns were witnessed in the case of Vedanta, Indian Hotels, Tata Communications, Coforge, Birla Soft, Hindalco and MRF**.** Similarly,"H_{02}:There was no significant impact of post-T +0 settlement on trading volumes"was rejected since there was a noticeable increase in the volumes post-T +0 settlement.

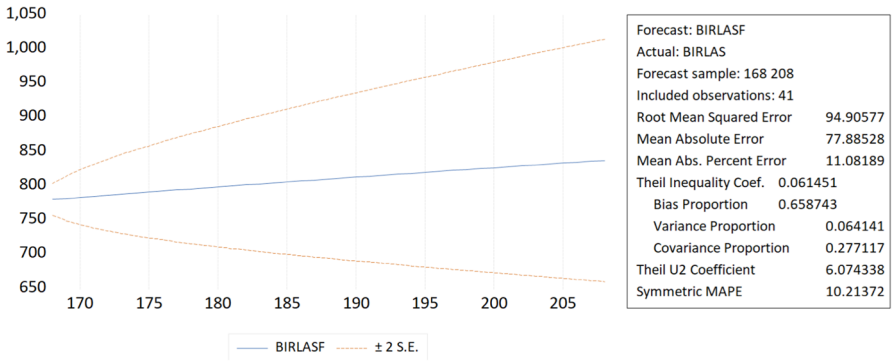

Fig. 4 Graphical representation of Dynamic Forecasting in the case of Birla Soft. Source: Author's computation/compilation (obtained from EViews)

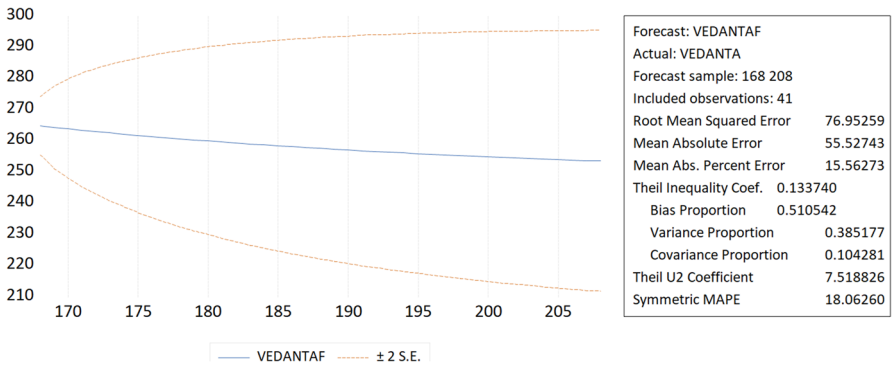

Fig. 5 Graphical representation of Dynamic Forecasting in the case of Vedanta. Source: Author's computation/compilation (obtained from EViews)

5 Discussion and Implications

Unravelling the impact of the T + 0 settlement cycle on Indian stock markets, it was observed that while introducing T + 0 resulted in an overall increase in trading volume, its effect on individual stock returns was mixed. Some stocks experienced significant gains, while others suffered losses. Statistical analysis reveals no significant average impact of T + 0 on stock returns across different timeframes. The mixed results show the need for further research into stock-specific factors and investor behaviour. However, the observed trading volume, which increased after the T + 0 settlement, promotes that the markets have accepted the T + 0 settlement cycle. The shortened settlement cycle has led institutional investors to adjust their investment patterns. FIIs show a shift in buying behaviour and DIIs face challenges in aligning liquidity with shorter settlement times, impacting overall market returns and sectoral investments (Singh et al., 2024).

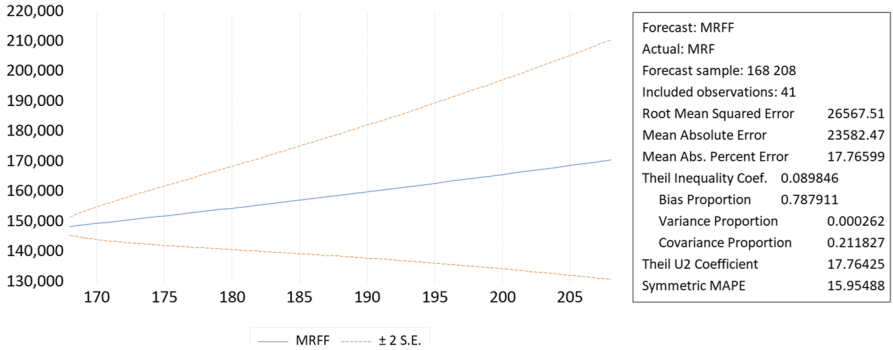

Fig. 6 Graphical representation of Dynamic Forecasting in the case of MRF. Source: Author's computation/compilation (obtained from EViews)

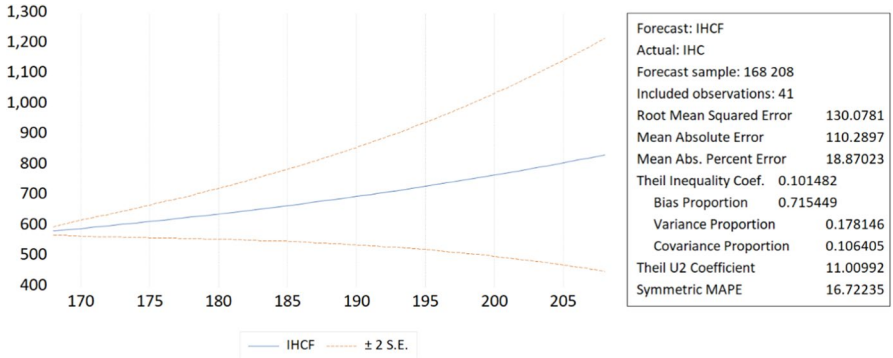

Fig. 7 Graphical representation of Dynamic Forecasting in the case of IHC. Source: Author's computation/compilation (obtained from EViews)

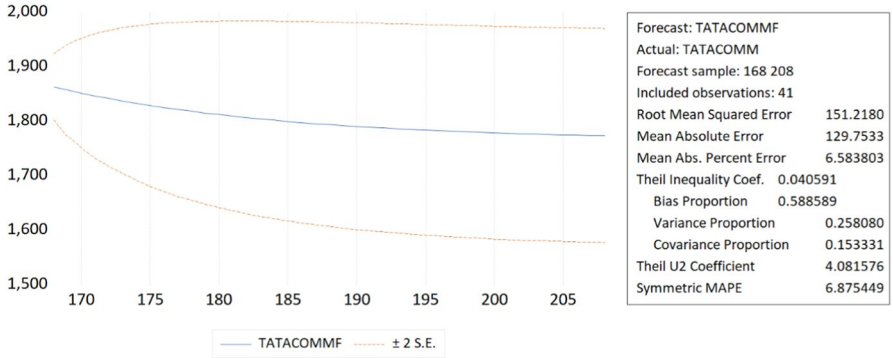

Fig. 8 Graphical representation of Dynamic Forecasting in the case of Tata Comm. Source: Author's computation/compilation (obtained from EViews)

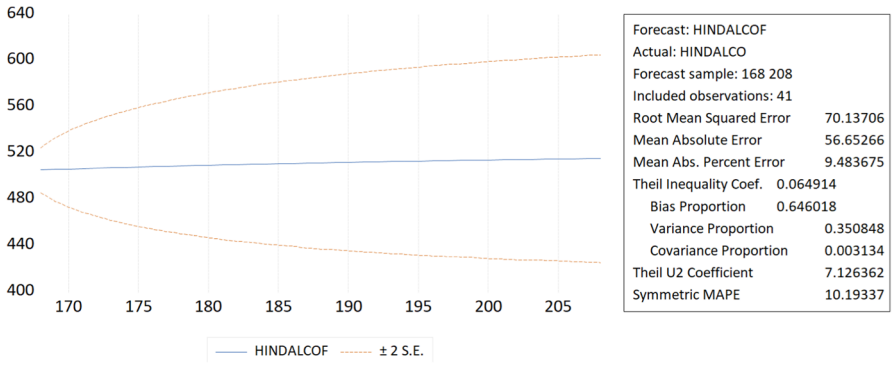

Fig. 9 Graphical representation of Dynamic Forecasting in the case of Hindalco. Source: Author's computation/compilation (obtained from EViews)

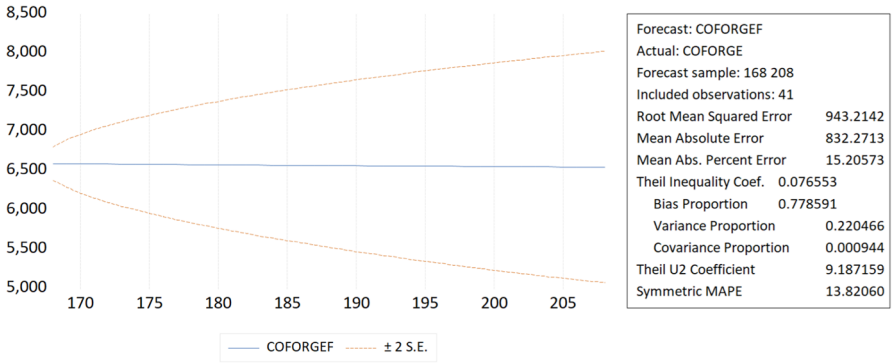

Fig. 10 Graphical representation of Dynamic Forecasting in the case of Coforge. Source: Author's computation/compilation (obtained from EViews)

5.1 Practical Implications

T + 0 settlement, where trades are settled on the same day as execution, offers both benefits and challenges for the Indian market. Supporters believe streamlining of the processes, reduce counterparty risk, and enhance intraday trading and liquidity. However, implementing T+ 0 settlement presents challenges, including technological upgrades, global coordination across time zones, and adjustments to IT systems, business models, and risk management frameworks to support near-instant settlement.

The Indian market's T+ 0 settlement beta launch presents a unique opportunity for global financial centres like the US, Europe, and Canada to learn from its outcomes. By examining India's experience, these regions can gain valuable insights into the regulatory, technological, and operational changes required to successfully implement ultra-short settlement cycles. Understanding the impact on market

participants, liquidity, and overall market dynamics in the Indian context can inform the decision-making process for adopting similar reforms in their respective jurisdictions.

T +0 settlement will help to increase the trading volume. As a result, there will be enhanced market liquidity and lower transaction cost. Furthermore, especially in cases of difficult- to-borrow stocks due to lower trading volumes resulting from high security prices, increased liquidity may be observed (Baig et al., 2022). This shorter cycle will help make policies for the regulatory body. Brokers usually generate significant income from customer by charging interest to them. But with shortened cycle they need to re-evaluate their business strategy. This will likely become the new standard for the stockbroker community (kaur et al., 2023).

Market participants like stock exchanges, brokers and depository participants need to invest more into the market infrastructure institution to offer T+ 0 settlement cycle in terms of resource allocation. As a result, small brokers need to consolidate with larger players to offer better products to their customers. Also, the closure or acquisition of small firms would result in the unemployment of small brokerage firms'employees.

5.2 Regulatory Implications and Coordination

Shortening settlement cycles requires collaboration among market participants, regulators, and infrastructure providers. Although the industry may independently reduce settlement cycles, a coordinated approach is essential for a smooth transition and preserving market integrity. ESMA's consultation on this issue shows the regulator's commitment to gathering input from stakeholders, including market infrastructure firms, investment firms, issuers, and investors, to assess the costs, benefits, and challenges of shorter settlement cycles.

Regulatory action might be needed to achieve a coordinated reduction in settlement cycles across the EU. ESMA's final report to the European Commission, expected in the fourth quarter of 2024, will offer recommendations based on consultation feedback and address any urgent issues stemming from North America's move to T+ 1 settlement (kaur et al., 2023).

5.3 The Role of Technology in Streamlining Settlement Cycles

Technological advancements are key to enabling shorter settlement cycles. As these cycles shorten, automation and digitization will be essential for efficient trade processing and coordination among market participants. Emerging technologies like distributed ledger technology (DLT) and blockchain offer potential to improve transparency, speed up settlements, and reduce operational risks. To maximize these benefits, market participants need to upgrade IT infrastructure, embrace digital solutions, and adopt standardized protocols. Automated trade confirmation, reconciliation, and reporting systems, along with real-time data and analytics, will help accelerate cycle times and enhance accuracy and risk management.

5.4 Market Impact and Considerations

The shift to shorter settlement cycles in Europe, whether T+ 1 or T+ 0, will significantly impact market participants. While benefits like reduced counterparty risk, increased capital efficiency, and improved market liquidity are expected, participants must also address the challenges. Operational adjustments will be crucial, requiring an evaluation of IT systems, workflow processes, and risk management frameworks to ensure smooth integration with the new settlement cycle.

Collaboration among market infrastructure firms, investment firms, and regulators is key to ensuring industry-wide readiness and addressing potential bottlenecks. Market participants must also evaluate the impact of shorter settlement cycles on liquidity management, collateral requirements, and funding arrangements. A thorough assessment of financial and operational implications will help develop robust strategies to adapt effectively to the changing landscape. Foreign Institutional Investors may face challenges with the T+ 0 reduced settlement cycle due to time zone differences. These overseas fund houses must allocate capital before trading and hedge for currency risks.

According to this study's findings, the Oil and Gas, retail, and food and beverage industries witnessed high average volumes. SEBI can increase the margins of these companies to prevent unnecessary speculation.

Alternative investment funds, portfolio managers, and fund managers handling mutual fund asset management portfolios or portfolio management schemes, as well as hedge funds, must realign their portfolios for companies experiencing unusually high trading volumes. However, there will be a need for investor education at a higher level since these measures will help increase investor confidence by reducing the counterparty risk.

5.5 Future Scope of Research

Future research can focus on the enduring effects of the T+ 0 settlement on market competence once it is fully implemented. Currently, it has been offered only for 25 securities, and that too on an opt-in basis. This implies that all stock brokers do not need to go for T+ 0 mandatorily since only a beta version of T+ 0 has been introduced. A detailed study of the behavioural facets of investors in reply to the T+ 0 settlement can be undertaken, focusing on sentiment-driven factors influencing market reactions.

6 Conclusion

T +0 settlement launch in the Indian equity markets has gradually impacted market dynamics by analysing 25 selected stocks. Although currently optional, the recent decision by SEBI has already demonstrated its potential to enhance market efficiency and liquidity. Our research indicates a varied impact on different stocks, suggesting that the effect of T+ 0 settlement is not consistent. Certain companies,

such as Vedanta, Indian Hotels, Tata Communications and Hindalco, saw significant positive abnormal returns, indicating an immediate favourable response to the shortened settlement cycle. Vedanta, in particular, consistently exhibited a strong positive trend across all event windows, resulting in a substantial 36% abnormal return over 20 days. However, the impact was not universally positive. Companies like Coforge, Birla Soft, and MRF experienced significant negative abnormal returns, especially in the longer event windows. This outcome disparity suggests that the benefits of T+ 0 settlement may be sector-specific or dependent on individual stock characteristics. Interestingly, while individual stocks showed significant movements, the cumulative average abnormal returns (CAAR) across all sample stocks were insignificant. The results were supported by dynamic forecasting, which showed significant differences between forecasted values and closing values for 6 sample stocks during the period of 27–02–2024 to 30–04–2024. This was confirmed through RMSE, TIEL, and bias proportion.T +0 settlement has had noticeable effects on certain stocks at this early stage, but its overall market impact remains subtle. The volume analysis provides encouraging signs for the potential of T+ 0 settlement. An average increase of 3% in trading volume across the participating stocks, even with the optional nature of the current implementation, suggests that market participants are beginning to embrace this new settlement cycle. The trivial increase in trading volume is a constructive indicator showing scope for further improvements once the T+ 0 settlement is implemented fully. These findings are important for the market regulator. Although the initial response to improved market dynamics is hopeful, the fluctuating impact on different stocks underscores the need for careful monitoring and necessary policy adjustments. Investors and traders must gauge stock-specific and industry-specific reactions to the T+ 0 settlement. Since SEBI will shortly make the T+ 0 settlement compulsory for all stocks, the findings of this study provide important input for improvising the future implementation strategy. Future research should focus on long-term impacts and sector-specific analyses to fully gauge the inferences of this significant change in market structure. Overall, the results of this study align with the semi-strong form of market efficiency, as the implementation of the T+ 0 settlement initiative demonstrates the market's ability to incorporate and reflect publicly available information swiftly. Early indicators suggest that this bold step is positively influencing the efficiency and dynamism of the Indian equity markets.

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